Earth-Observation Data Processing: Second report - ISM Test

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In this part, the Level-1B radiances were orthorectified by projecting them from the instrument's irregular geometry onto a regular Military Grid Reference System (MGRS) grid oriented North-South. This is achieved by creating a 2D interpolant of the L1B geodetic coordinates and radiances, then extracting unique MGRS tiles corresponding to a 100-meter precision level. Finally, the function evaluates the interpolant at these new regularized coordinates to produce the L1C radiances.

Figure 1 visualizes this transformation by outlining the original L1B footprint in blue and mapping the new L1C MGRS coordinates as red dots. The L1B grid is visibly inclined because its spatial sampling is strictly tied to the satellite sensor's specific acquisition geometry. In contrast, the red L1C dots form a North-South aligned grid of dots, which provides the common reference frame necessary for stitching images and co-registering different spectral bands.

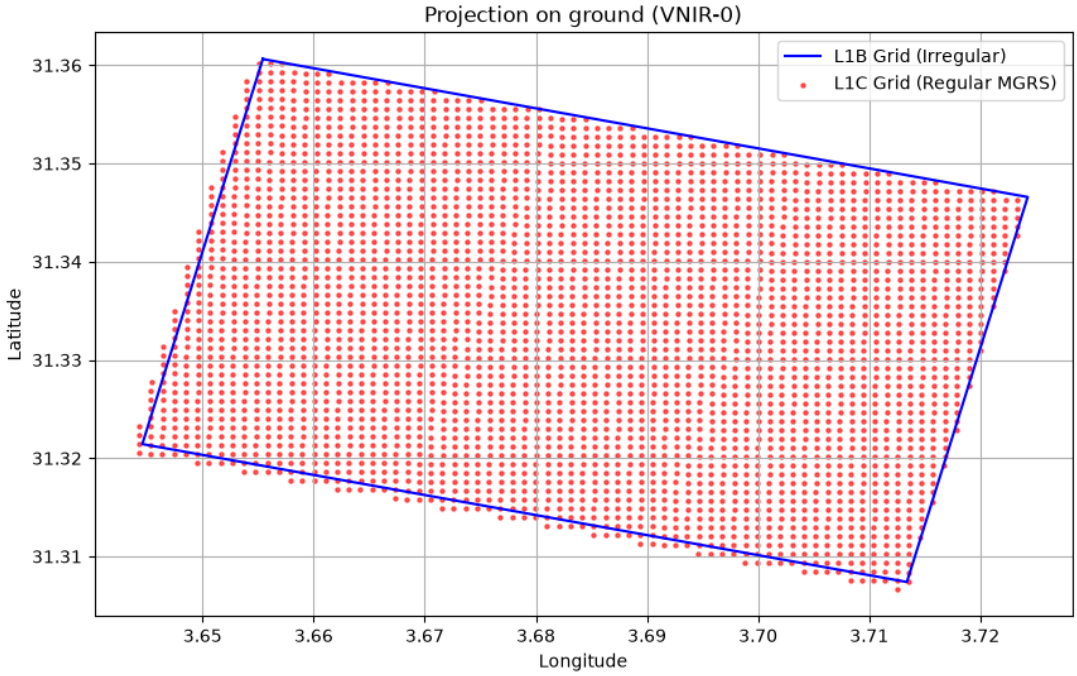


Figure 1: L1B (irregular) and L1C (regular) grids.

The spatial sampling distance plot (Figure 2) illustrates that the physical ground distance between consecutive pixels is not constant across the sensor's swath. Calculated

along the central row of the L1B geometry, the pixel-to-pixel distance fluctuates slightly, which aligns directly with the sensor's baseline Spatial Sampling Distance. These minor variations across the track are a natural geometric result of the instrument's changing viewing angles across the focal plane before orthorectification is applied.

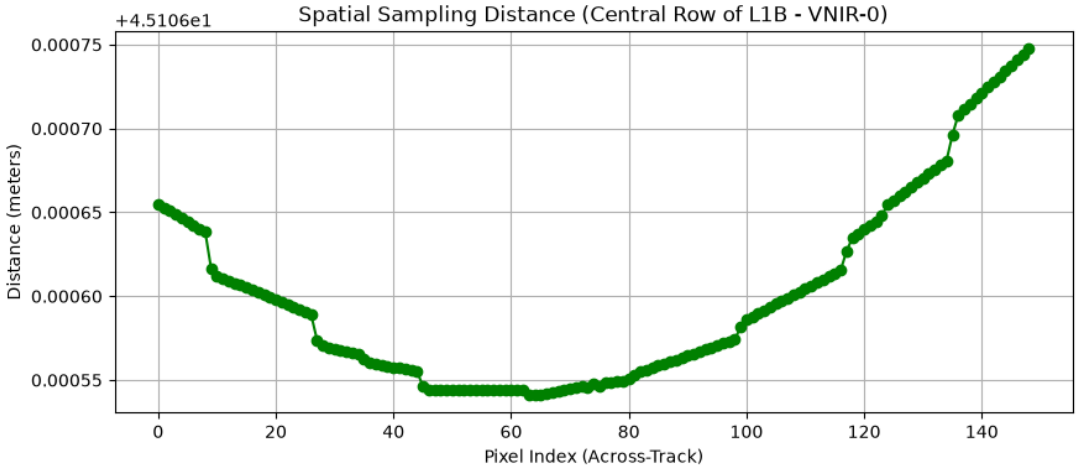


Figure 2: Spatial Sampling Distance.